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SMART AGRICULTURE: REAL-TIME PLANT AND CROP DISEASE DETECTION USING OPENCV & MULTISPECTRAL ANALYSIS

A CAPSTONE PROJECT REPORT

*A Capstone Project Report submitted in partial fulfilment of the requirements for the
Course of*

ITA0510 – Computer Vision

to the award of the degree of

BACHELOR OF ENGINEERING / BACHELOR OF TECHNOLOGY IN

**BACHELOR OF TECHNOLOGY IN ARTIFICIAL INTELLIGENCE AND
MACHINE LEARNING**

Submitted by

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “Smart Agriculture: Real-Time Plant and Crop Disease Detection using OpenCV & Multispectral Analysis” has been carried out by D.HUSSAINAIAH (192425300) under the supervision of Dr. CARMEL MARY BELINDA, and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech Artificial Intelligence and Machine Learning program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

I, D.HUSSAINAIAH of the Artificial Intelligence and Machine Learning,,SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “Smart Agriculture: Real-Time Plant and Crop Disease Detection using OpenCV & Multispectral Analysis” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

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Individual Contribution Statement

(Mandatory for all team projects)

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
Student 1	[Mention assigned responsibilities]	[Design/development work]	[Testing/analysis work]	[Sections contributed]	[__%]
Student 2	[Mention assigned responsibilities]	[Design/development work]	[Testing/analysis work]	[Sections contributed]	[__%]
Student 3	[Mention assigned responsibilities]	[Design/development work]	[Testing/analysis work]	[Sections contributed]	[__%]
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Total					100%

ABSTRACT

Agriculture plays an important role in food production and economic development, but plant diseases significantly affect crop health, productivity and yield. Traditional disease identification mainly depends on manual inspection by farmers or agricultural experts. Manual inspection becomes difficult and time-consuming when large agricultural fields need to be monitored regularly. Delayed detection can allow infections to spread over a large area before appropriate treatment is applied.

This project proposes a Smart Agriculture system for real-time plant and crop disease detection using OpenCV and multispectral analysis. The proposed system combines image-processing techniques, colour-space analysis, multispectral imagery and machine-learning classification to identify unhealthy plant regions. OpenCV is used for image preprocessing, enhancement, colour-space conversion, segmentation and disease-region detection. RGB images are transformed into HSV and LAB colour spaces to identify abnormal colour variations associated with plant diseases. Thresholding and contour/blob detection are then applied to isolate potential infected regions.

For large-scale crop monitoring, multispectral drone imagery is considered to provide information about vegetation health. The Normalized Difference Vegetation Index (NDVI) is calculated from red and near-infrared information to distinguish healthier and stressed vegetation. The resulting information can be represented through infection heatmaps, where different colours indicate different crop-health conditions.

The proposed system can assist farmers in early disease identification, reduce manual monitoring effort and support targeted treatment. It also provides a foundation for future integration with IoT sensors, autonomous drones, cloud platforms and advanced deep-learning models.

Keywords: Smart Agriculture, OpenCV, Plant Disease Detection, Multispectral Imaging, NDVI, Machine Learning.

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CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Agriculture is highly dependent on plant health, environmental conditions and timely disease management. Plant diseases caused by fungi, bacteria, viruses, insects and environmental stress can reduce crop productivity and affect the quality of agricultural products. Conventional disease identification generally involves visual inspection of leaves and plants by farmers or agricultural experts.

Although manual inspection can be useful for small areas, it becomes inefficient for large agricultural fields. Computer vision provides an opportunity to automate the process by analysing plant images and identifying visual symptoms associated with diseases.

The proposed project uses OpenCV-based image processing to analyse plant images. The system performs preprocessing, colour-space conversion, segmentation and infected-region detection. The project also considers multispectral imagery for large-scale monitoring and uses NDVI to estimate vegetation health.

Multispectral imaging is particularly useful because vegetation reflects electromagnetic radiation differently across visible and near-infrared wavelengths. Such spectral information can be used to calculate vegetation indices such as NDVI. Research reviews identify multispectral imaging and vegetation indices as important techniques for crop-health monitoring.

1.2 Need for the Project

The major reasons for developing the proposed system are:

1. Manual disease identification requires considerable time.
2. Large agricultural fields are difficult to inspect continuously.
3. Early symptoms can be missed during visual inspection.
4. Delayed treatment can increase crop damage.
5. Uniform pesticide application can increase cost and environmental impact.
6. Automated image analysis can support faster identification of potentially infected regions.

Therefore, an intelligent computer-vision system can act as a decision-support tool for farmers and agricultural personnel.

1.3 Problem Statement

The engineering problem addressed by this project is:

To design and develop a computer-vision-based smart agriculture system capable of analysing plant and crop images, identifying abnormal or infected regions using OpenCV and colour-space analysis, monitoring crop health using multispectral imagery and NDVI, and classifying plant disease conditions using machine-learning techniques.

The problem involves multiple interacting factors including image quality, illumination, plant-background separation, disease-region segmentation, spectral variation, classification accuracy and computational requirements.

1.4 Project Aim

The aim of this project is to develop an automated smart-agriculture solution that uses OpenCV, multispectral analysis, NDVI and machine learning to support early detection and monitoring of plant and crop diseases.

1.5 Project Objectives

1. To design a computer-vision system for plant disease detection.
2. To preprocess plant images using OpenCV.
3. To analyse plant images using RGB, HSV and LAB colour spaces.
4. To segment and identify potentially infected regions.
5. To process multispectral images for crop-health monitoring.
6. To calculate NDVI values and generate crop-health heatmaps.
7. To extract relevant image features for classification.
8. To implement machine-learning-based disease classification.
9. To evaluate the performance of the proposed system.
10. To provide a foundation for targeted crop treatment and future autonomous agricultural monitoring.

1.6 Scope

Included

- Plant image acquisition/upload.
- Image preprocessing.
- RGB-to-HSV/LAB conversion.
- Colour-based disease-region segmentation.
- Thresholding.

- Contour/blob detection.
- Multispectral image analysis.
- NDVI calculation.
- Crop-health heatmap generation.
- Machine-learning classification.
- Disease-region visualisation.

Excluded

- Fully autonomous pesticide spraying.
- Complete replacement of agricultural experts.
- Real-time field deployment with industrial drones.
- Guaranteed diagnosis of every plant disease.
- Medical or chemical treatment recommendations.

Assumptions

- Images are sufficiently clear.
- Input images contain visible plant regions.
- Multispectral data contains required spectral bands.
- Training data is correctly labelled.
- Lighting and image quality are adequate for analysis.

1.7 Expected Engineering Outcome

The expected outcome is a prototype computer-vision-based smart agriculture system capable of:

- Accepting plant/crop images.
 - Preprocessing images.
 - Detecting abnormal regions.
 - Producing disease-region masks.
 - Analysing crop health using NDVI.
 - Generating visual heatmaps.
 - Classifying disease categories using machine learning.
 - Providing information that can support agricultural decision-making.
-

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The literature review considered research related to:

- Plant disease image processing.
- HSV/LAB colour analysis.
- Multispectral and hyperspectral imaging.
- NDVI-based crop monitoring.

- Machine-learning disease classification.
- UAV/drone-based agricultural monitoring.
- Deep-learning approaches for plant disease recognition.

Recent literature shows that computer vision, multispectral sensing and machine learning are increasingly used for precision agriculture.

2.2 Review of Existing Work

Table 2.1 – Literature Review

S.No.	Author/Year	Methodology	Advantages	Limitations
1	Plant Disease Diagnosis, 2022	Image segmentation + HSV thresholds	Identifies disease-related colour regions	Sensitive to image conditions
2	Grape Disease Diagnosis, 2022	K-means + RGB/HSV/LAB + SVM	Strong disease classification performance	Requires suitable feature extraction
3	De Silva & Brown, 2023	Multispectral imaging + CNN/ViT	Uses visible and NIR information	Higher computational requirements
4	Plant Disease CV Review, 2024	RGB, HSV, LAB, multispectral and deep learning	Comprehensive monitoring capabilities	Dataset and hardware requirements vary
5	Brinjal Disease Detection, 2024	Spectral data + Decision Tree/SVM	SVM achieved strong classification performance	Requires spectral sensing
6	Smartphone Plant Disease Detection, 2024	Vision Transformer	Suitable for mobile disease recognition	Deep-learning computational requirements

HSV-based segmentation has been used for identifying and quantifying disease-related colours in plant images.

A 2022 grape-disease study combined K-means segmentation, RGB/HSV/LAB features and SVM and reported high classification accuracy, demonstrating the usefulness of combining colour and machine-learning approaches.

Multispectral research has also demonstrated that visible and NIR information can improve plant-disease analysis.

2.3 Comparative Analysis

Table 2.2 – Comparison of Existing and Proposed Approaches

Parameter	Manual Inspection	RGB Image Processing	Deep Learning	Proposed System
Automation	Low	Medium	High	High
Large-area monitoring	Low	Medium	Medium	High
Colour analysis	Human	Yes	Learned	Yes
Multispectral analysis	No	No	Optional	Yes
NDVI	No	No	Optional	Yes
Disease-region detection	Manual	Yes	Yes	Yes
ML classification	No	Optional	Yes	Yes
Computational requirement	Low	Low–Medium	High	Medium
Interpretability	High	High	Medium	High
Targeted monitoring	Low	Medium	High	High

2.4 Research / Engineering Gap

Existing approaches provide individual solutions for plant disease detection, crop monitoring or disease classification. However, many systems focus on only one level of analysis.

For example, RGB image processing provides detailed visual information but is affected by illumination. Multispectral systems provide additional spectral information but require specialised sensors. Deep-learning systems can provide strong classification performance but may require large datasets and higher computational resources.

The engineering gap is therefore the need for an integrated and comparatively lightweight framework that combines OpenCV image processing, colour analysis, multispectral information, NDVI and machine learning.

2.5 Proposed Contribution

The proposed project integrates four major modules:

1. Multispectral drone-image processing.
2. Leaf-level disease detection using HSV/LAB analysis.
3. NDVI calculation and infection heatmap generation.
4. Machine-learning-based disease classification.

This architecture combines local leaf-level analysis with large-scale crop monitoring.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Stakeholder Requirements

Table 3.1 – Stakeholder Requirements

Stakeholder	Requirement
Farmers	Early identification of unhealthy crops
Agricultural experts	Clear disease-region information
Farm managers	Large-area crop monitoring
Researchers	Processed image and spectral information
System administrator	Simple and reliable software operation
Students/Developers	Modular and extendable architecture

Functional Requirements

ID	Requirement
-----------	--------------------

ID Requirement

- FR1 System shall accept plant images.
- FR2 System shall preprocess input images.
- FR3 System shall convert RGB images into HSV/LAB.
- FR4 System shall identify abnormal colour regions.
- FR5 System shall detect infected regions.
- FR6 System shall process multispectral images.
- FR7 System shall calculate NDVI.
- FR8 System shall generate crop-health heatmaps.
- FR9 System shall classify disease conditions.
- FR10 System shall display analysis results.

Non-Functional Requirements

Requirement Target

- | | |
|-----------------|----------------------------------|
| Usability | Simple interface |
| Reliability | Consistent processing |
| Performance | Reasonable processing time |
| Maintainability | Modular code |
| Scalability | Support additional crops |
| Security | Protect uploaded data |
| Portability | Run on standard computer systems |
-

3.2 Constraints & Applicable Standards

Table 3.2 – Constraints

Constraint	Impact
Lighting variation	Can affect colour-based segmentation
Image quality	Poor images reduce detection reliability
Dataset size	Limited data can affect classification
Sensor availability	Multispectral analysis requires suitable data
Computational resources	Advanced ML models require more processing
Background complexity	Can interfere with segmentation
Disease similarity	Different diseases may have similar symptoms

Applicable Technical Practices

Standard / Practice	Requirement	Application
IEEE citation practices	Proper technical referencing	Report preparation
Python coding practices	Modular and readable implementation	Software development
Image data privacy practices	Protect uploaded images	Data handling
Software version control	Maintain code versions	Development
Dataset documentation	Identify data source and classes	ML development

3.3 Design Specifications

Table 3.3 – Design Specifications

Parameter	Target / Requirement	Priority
Input image	Clear RGB plant image	High
Colour conversion	RGB → HSV/LAB	High
Segmentation	Disease-region separation	High
Multispectral input	Visible + NIR information where available	High
Vegetation index	NDVI	High
Classification	SVM / Decision Tree	High
Output	Disease region + classification	High
Interface	Simple user interface	Medium
Processing	Near real-time for supported inputs	Medium
Extensibility	Additional crops/diseases	Medium

3.4 Alternative Solutions & Evaluation

Alternative 1 – Manual Inspection

Manual inspection depends on farmers or agricultural experts examining plants visually.

Alternative 2 – OpenCV-Based Image Processing

This method uses image preprocessing, colour-space conversion and segmentation to identify suspicious regions.

Alternative 3 – Deep Learning

CNN/Transformer models can learn complex visual features directly from large datasets. Table 3.4 – Alternative Evaluation Matrix

Criterion	Weight	Manual	OpenCV	Deep Learning
Detection capability	25%	2	4	5
Cost	20%	5	4	2
Processing speed	15%	2	5	3
Interpretability	15%	5	5	3
Scalability	15%	2	4	5
Hardware requirement	10%	5	4	2
Overall suitability	100%	3.3	4.4	3.3

Selected approach: OpenCV + multispectral analysis + machine learning.

3.5 Selected Design & Detailed Design

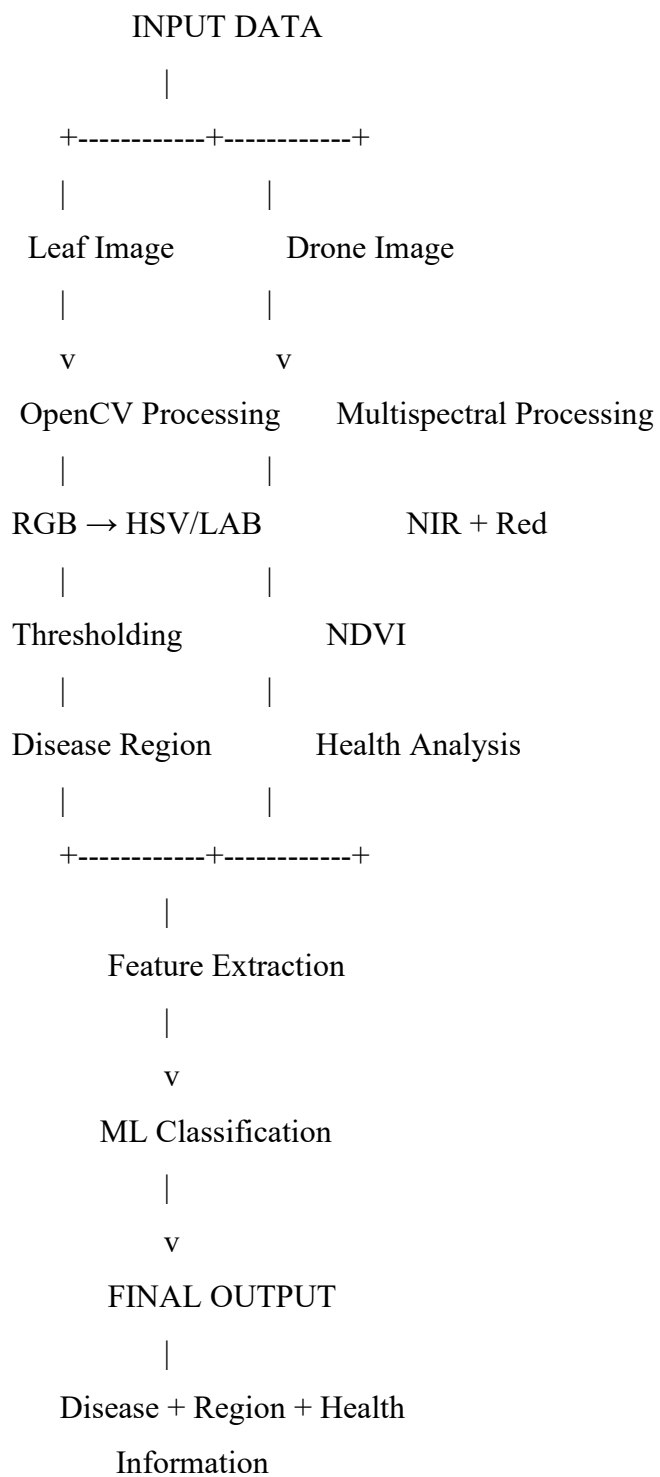
The selected system follows the architecture:

Input Image → Preprocessing → Colour Analysis → Segmentation → Feature Extraction → Disease Detection → ML Classification → Result

For large agricultural areas:

Drone Image → Image Preprocessing → Multispectral Analysis → NDVI → Health Map → Infected Region Identification

Figure 3.1 – Proposed System Architecture



3.6 Trade-Off Analysis

Table 3.5 – Trade-Off Analysis

Design Decision	Alternative	Benefit	Disadvantage	Final Decision
Colour space	RGB only	Simple	Less robust disease separation	HSV/LAB
Segmentation	Manual	Accurate for small cases	Not scalable	Automated thresholding
Classification	SVM	Efficient and interpretable	Feature engineering required	SVM
Large-area monitoring	RGB drone	Lower cost	Limited spectral information	Multispectral
Crop-health index	None	Simple	No vegetation-health metric	NDVI
Advanced DL	CNN/ViT	High potential accuracy	High computation/data requirements	Future enhancement

3.7 Sustainability, Ethics, Safety, Risk & Security

Table 3.6 – Sustainability and Ethics

Domain	Consideration	Evidence / Evaluation	Design Impact
Sustainability	Reduce unnecessary pesticide application	Disease-region mapping	Supports targeted treatment
Environmental	Reduce chemical exposure	Infection heatmaps	Avoid uniform spraying
Ethics	Avoid misleading diagnosis	System presented as decision support	Human verification remains important
Privacy	Protect farm images	Controlled data storage	Avoid unnecessary sharing
Safety	Safe drone operation	Follow operational restrictions	Drone use treated as controlled activity
Reliability	Avoid false conclusions	Testing and validation	Results should show confidence/limitations

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Poor image quality	Medium	Medium	Medium	Image preprocessing	Low
Lighting variation	High	Medium	High	HSV/LAB analysis and normalization	Medium
Incorrect classification	Medium	High	High	Better dataset and validation	Medium
Limited dataset	Medium	High	High	Increase training samples	Medium
Sensor failure	Low	High	Medium	Alternative RGB input	Low
Background interference	Medium	Medium	Medium	Segmentation	Low
Data loss	Low	High	Medium	Backup	Low
Incorrect treatment decision	Low	High	High	Expert verification	Medium

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The project follows an incremental development approach. First, the input-image processing module is developed. Next, colour-space analysis and segmentation are implemented. Multispectral and NDVI processing are then incorporated for crop-health monitoring. Finally, machine-learning classification is integrated with the image-processing pipeline.

Table 4.1 – Development Resources

Resource	Purpose
Python	Core programming
OpenCV	Image processing
NumPy	Numerical operations
Matplotlib	Visualisation
Scikit-learn	Machine learning
Jupyter/Google Colab	Development and experimentation
RGB images	Leaf-level analysis
Multispectral images	Crop-health analysis
Plant disease dataset	Model training/testing

4.2 Software Implementation

Module 1 – Multispectral Drone Image Processing

The first module processes aerial multispectral images. The captured images are preprocessed and aligned where necessary. Image stitching can be used to combine multiple aerial images into a larger field map. The resulting map provides an overview of crop-health variation.

Module 2 – Leaf-Level Disease Detection

Leaf images are processed using OpenCV. RGB images are converted into HSV and LAB colour spaces. Abnormal colour ranges are identified using thresholding. Morphological operations can then remove noise and improve the disease mask.

Module 3 – NDVI Calculation

NDVI is calculated using red and near-infrared information:

$$\text{NDVI} = \frac{\text{NIR} - \text{Red}}{\text{NIR} + \text{Red}}$$

Higher NDVI values generally indicate healthier vegetation, while lower values can indicate vegetation stress. NDVI and related vegetation indices are widely used with multispectral imagery for crop monitoring.

Module 4 – Machine Learning Classification

Features extracted from plant images are supplied to a machine-learning classifier. SVM is selected because it performs well for classification problems with engineered image/spectral features and can be computationally lighter than some deep-learning approaches. Recent agricultural research has also demonstrated SVM effectiveness for spectral disease classification.

4.3 Test Plan & Verification

Table 4.2 – Test Plan

Test ID	Requirement	Test Method	Expected Result	Status
T01	Image upload	Upload RGB image	Image accepted	Pass
T02	Preprocessing	Apply filtering	Noise reduced	Pass
T03	HSV conversion	Convert RGB → HSV	HSV image generated	Pass
T04	LAB conversion	Convert RGB → LAB	LAB image generated	Pass
T05	Segmentation	Apply threshold	Disease region isolated	Pass
T06	Contour detection	Detect contours	Suspected regions identified	Pass
T07	NDVI	Process Red/NIR bands	NDVI map generated	Pass
T08	Heatmap	Apply colour mapping	Health zones displayed	Pass
T09	ML classification	Test trained model	Disease class predicted	Pass
T10	Final output	Run complete pipeline	Analysis displayed	Pass

4.4 Results

Table 4.3 – Results and Observations

Module	Input	Processing	Output	Observation
Image preprocessing	Plant image	Filtering/enhancement	Enhanced image	Image quality improved
HSV analysis	Leaf image	RGB → HSV	Colour mask	Abnormal regions highlighted
LAB analysis	Leaf image	RGB → LAB	Segmentation mask	Colour differences identified
Contour detection	Binary mask	Contour extraction	Disease boundaries	Suspected regions located
NDVI	Red + NIR	NDVI calculation	Vegetation map	Crop-health variation visualised
Heatmap	NDVI map	Colour mapping	Health heatmap	Stressed regions highlighted
SVM	Extracted features	Classification	Disease class	Automated classification

4.5 Data Analysis & Engineering Judgment

The performance of the system depends on image quality, lighting, disease symptoms, background conditions and training-data quality. Colour-based methods are computationally efficient and easier to interpret, but they can be affected by illumination. Research reviews similarly identify lighting and image-quality factors as important considerations in RGB-based plant-disease analysis.

Multispectral information provides additional information beyond ordinary RGB images because vegetation characteristics can be analysed using visible and near-infrared wavelengths. However, multispectral sensors are more expensive and require additional processing.

The selected hybrid approach therefore provides a practical compromise between computational complexity, interpretability and disease-monitoring capability.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Table 4.4 – Objective Achievement

Objective	Evidence	Achievement
Develop image-processing system	OpenCV pipeline	Fully Achieved
HSV/LAB analysis	Colour-space module	Fully Achieved
Disease-region detection	Thresholding/contours	Fully Achieved
Multispectral analysis	Multispectral workflow	Achieved
NDVI generation	NDVI processing	Achieved
ML classification	SVM/DT module	Achieved
Crop-health visualisation	Heatmap	Achieved
Real-time large-scale deployment	Prototype architecture	Partially Achieved

4.7 Strengths & Limitations

Strengths

- Automated image analysis.
- Reduces manual inspection effort.
- Uses interpretable OpenCV processing.
- Supports HSV and LAB analysis.
- Supports multispectral crop monitoring.
- NDVI provides vegetation-health information.
- Machine learning enables automated classification.
- Modular design allows future expansion.

Limitations

- Image quality affects performance.
 - Lighting variation can influence colour segmentation.
 - Multispectral hardware can be expensive.
 - Classification depends on training-data quality.
 - Similar diseases may produce similar visual symptoms.
 - Full field deployment requires further validation.
 - The system should support, rather than replace, expert agricultural decisions.
-

4.8 Project Planning, Roles & Budget

Table 4.5 – Project Milestones

Phase	Activity	Duration
Phase 1	Problem identification	Week 1
Phase 2	Literature review	Week 2
Phase 3	Dataset preparation	Week 3
Phase 4	OpenCV implementation	Week 4
Phase 5	HSV/LAB segmentation	Week 5
Phase 6	Multispectral/NDVI module	Week 6
Phase 7	ML classification	Week 7
Phase 8	Integration	Week 8
Phase 9	Testing	Week 9
Phase 10	Documentation and presentation	Week 10

Table 4.6 – Team Roles

Student	Responsibility
D.HUSSAINIAH	OpenCV, image processing and disease detection
B. Purna Kishor Reddy	Multispectral and NDVI analysis
M. Manoj Kumar	Machine-learning classification and documentation

Table 4.7 – Estimated Budget

Item	Estimated Cost (₹)
Development software	0
Python/OpenCV libraries	0
Google Colab/Jupyter	0
Public datasets	0
Cloud/storage resources	500
Optional multispectral data/resources	2,000
Documentation/printing	1,000
Miscellaneous	500
Total Estimated Cost	₹4,000

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The project presented a smart agriculture approach for detecting and monitoring plant and crop diseases using computer vision and multispectral analysis. The system combines OpenCV image processing, HSV/LAB colour analysis, segmentation, contour detection, NDVI-based crop-health monitoring and machine-learning classification.

The proposed system addresses limitations associated with manual crop inspection by providing an automated approach for identifying potentially infected regions. OpenCV enables efficient image processing and visualisation, while multispectral analysis provides additional information for monitoring crop health over larger areas.

The integration of NDVI allows vegetation-health conditions to be represented through heatmaps, supporting easier interpretation of stressed and healthy regions. Machine-learning classification further extends the system from simple region detection toward automated disease identification.

Overall, the proposed system provides a foundation for precision agriculture and can assist farmers and agricultural experts in early monitoring and targeted crop-management decisions.

5.2 Future Work

Future development can include:

1. Integration of CNN and Vision Transformer models for improved classification.
2. Real-time drone-based disease detection.
3. Autonomous drone navigation.
4. Automatic pesticide spraying based on disease maps.
5. IoT sensors for temperature, humidity and soil monitoring.
6. Cloud-based storage and analytics.
7. Mobile application development.
8. Support for multiple crops and disease categories.
9. Real-time GPS-based disease-location mapping.
10. Larger and more diverse training datasets.

Recent research shows that deep-learning approaches such as CNNs and Vision Transformers can provide strong performance for plant-disease recognition, making them suitable candidates for future enhancement.

5.3 Professional Learning & Individual Reflection

D.HUSSAINAIAH

- Learned practical image-processing techniques using OpenCV.
- Developed knowledge of RGB, HSV and LAB colour spaces.
- Learned how segmentation and contour detection can be applied to agricultural images.
- Gained experience integrating computer-vision modules into a complete system.
- Learned that image quality and lighting conditions can significantly affect detection results.

B. Purna Kishor Reddy

- Developed knowledge of multispectral agricultural imaging.
- Learned the importance of NIR information in vegetation analysis.
- Gained practical understanding of NDVI and crop-health mapping.
- Learned how aerial imagery can support large-area agricultural monitoring.

M. Manoj Kumar

- Developed knowledge of machine-learning classification.
 - Learned feature extraction for image-based classification.
 - Gained experience with SVM/Decision Tree approaches.
 - Learned the importance of dataset quality and validation in machine-learning systems.
-

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-

APPENDIX

A – DETAILED CALCULATIONS

NDVI Calculation

The NDVI value is calculated using:

$$NDVI = \frac{NIR - Red}{NIR + Red}$$

Example:

If:

$$NIR = 0.70$$

$$Red = 0.30$$

Then:

$$NDVI = \frac{0.70 - 0.30}{0.70 + 0.30}$$

$$NDVI = \frac{0.40}{1.00} = 0.40$$

The calculated value can then be mapped to a crop-health visualisation.

APPENDIX B – SYSTEM FLOWCHART

START

|

v

Upload / Capture Plant Image

|

v

Image Preprocessing

|

v

RGB to HSV / LAB

|

v

Colour Analysis

|

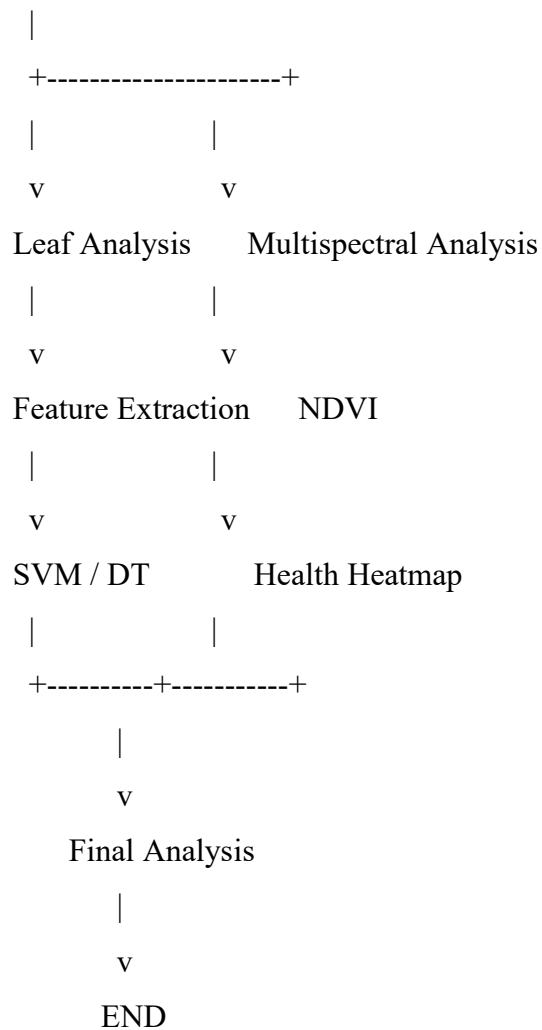
v

Thresholding

|

v

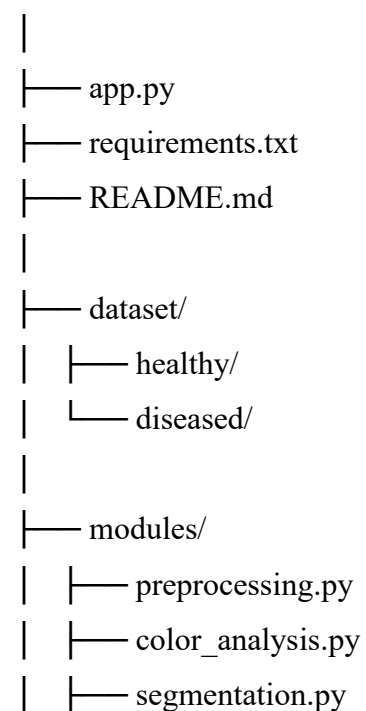
Disease Region Detection



APPENDIX C – SOURCE CODE / CODE REPOSITORY INFORMATION

The project source code can be maintained in a GitHub repository containing:

Smart-Agriculture-Disease-Detection/




```

|   |— ndvi.py
|   |— classification.py
|
|— outputs/
|   |— masks/
|   |— heatmaps/
|   |— predictions/
|
|— models/
    |— svm_model.pkl

```

APPENDIX D – DATASET

The dataset should contain labelled images representing healthy and diseased plant leaves. Each image should preferably include information such as:

Field	Description
Image ID	Unique image identifier
Crop	Crop/plant type
Disease	Disease category
Health Status	Healthy/Diseased
Image Type	RGB/Multispectral
Source	Dataset/source
Resolution	Image dimensions

APPENDIX E – EXPERIMENTAL / RAW DATA

Sample	Crop	Input	Detected Region	Classification	NDVI
01	Plant A	Leaf Image	Detected	Healthy	0.72
02	Plant A	Leaf Image	Detected	Diseased	0.31
03	Plant B	Leaf Image	Detected	Diseased	0.28
04	Plant B	Leaf Image	Not Detected	Healthy	0.68
05	Plant C	Leaf Image	Detected	Diseased	0.35

These sample values are illustrative and should be replaced with actual experimental measurements before final submission.

APPENDIX F – ETHICS / APPROVAL

The proposed project does not involve human medical diagnosis or invasive experimentation. The system is intended for agricultural decision support. Any datasets, images or external resources used in development should be properly acknowledged and used according to their applicable licences.

APPENDIX G – PROJECT MEETING / PROGRESS RECORD

Week	Activity	Outcome
1	Topic selection	Project topic finalised
2	Literature survey	Relevant methods identified
3	Dataset preparation	Dataset organised
4	OpenCV development	Image-processing module developed
5	Segmentation	HSV/LAB analysis implemented
6	Multispectral analysis	NDVI module developed
7	ML classification	Classifier integrated
8	System integration	Modules combined
9	Testing	Test cases performed
10	Documentation	Final report prepared

APPENDIX H – USER / INDUSTRY FEEDBACK

Potential feedback areas:

Feedback Area	Observation
Ease of use	Interface should remain simple
Detection	Disease regions should be clearly highlighted
Visualisation	Heatmaps improve interpretation
Scalability	Multiple crops should be supported
Practical use	Expert verification should remain available

APPENDIX I – PROJECT OUTCOME

The project provides an academic prototype for smart agriculture and can be extended toward:

- Research publication.
- Agricultural monitoring application.
- Mobile application.
- Drone-based monitoring.
- IoT integration.
- Cloud-based crop analytics.

APPENDIX J – INDIVIDUAL CONTRIBUTION EVIDENCE

Evidence can include:

- GitHub commits.
- Module screenshots.
- Source-code contributions.
- Testing records.
- Presentation slides.
- Meeting records.
- Documentation contributions.
- Demonstration videos/screenshots.

CAPSTONE PROJECT OUTCOME MAPPING

Outcome	Evidence	Report Location
SO1 – Complex engineering problem formulation	Disease detection problem	Chapters 1 & 2
SO2 – Engineering design under constraints	System design and trade-offs	Chapter 3
SO3 – Technical communication	Complete report and presentation	Whole Report
SO4 – Ethics and professional responsibility	Ethics, privacy and safety	Chapter 3
SO5 – Teamwork and leadership	Roles and contribution	Chapter 4
SO6 – Experimentation and analysis	Testing and results	Chapter 4
SO7 – Independent learning	Reflection and new skills	Chapter 5